Medical Insurance Claim Fraud Detection: A Reproducible Traditional Machine Learning Baseline for an Indian-Context Synthetic Study

B Varshith (23BDS011), M Jagadeshwar (23BDS033), J Ganesh (23BDS024)
Department of Data Science and AI, IIIT Dharwad, India
Faculty Adviser: Ramesh Athe

Abstract—This paper presents an interpretable traditional-machine-learning baseline for medical insurance claim fraud screening in an Indian context. The bundled workbook was preserved, audited and not used as the primary corpus because it contains only 4,500 records and lacks Indian policy, geography and historical-claim fields. A deterministic, explicitly synthetic INR-denominated claim population was generated with policy structures, regional/tier cost variation, allopathic and Ayurvedic treatment types, and probabilistic fraud labels. Train-only preprocessing, five-fold F2-led tuning, validation-only threshold selection and a held-out test comparison were applied to 15 classifiers. The best held-out synthetic result was XGBoost with F2=0.988, recall=0.993 and PR-AUC=0.999. Results are not estimates of real insurer effectiveness. The study contributes a transparent software and reporting baseline that includes class-imbalance comparison, INR-oriented screening-cost assumptions, calibration, fairness audit and interpretable feature assets. **Keywords—**health insurance fraud, imbalanced learning, India, explainable AI, F2 score, synthetic data.

I. INTRODUCTION

Medical insurance screening must distinguish suspicious patterns from legitimate, costly care while preserving fairness and due process. Indian products include individual, family-floater and group covers, government schemes, tiered provider costs and diverse treatment practices. A classifier therefore provides only one risk signal; a human reviewer must validate documents, medical necessity and policy clauses. This project establishes the transparent tabular baseline that later deep-learning and evidence-grounded agentic approaches can compare against.

II. RELATED WORK

Fraud detection literature stresses class imbalance, evolving behavioural patterns and careful evaluation [1]–[4]. Health-care studies use provider and claim features but face label and policy-context limitations [5], [6]. SMOTE and cost-sensitive learning address minority detection [7]–[9], while forests and boosted trees remain strong structured-data baselines [10]–[13]. Explainability and fairness work warns that predictive features do not constitute causal or legal evidence [17], [20], [21].

III. DATA AND METHODS

The raw workbook is archived with a checksum and a documented adequacy decision. The fallback generator creates right-skewed INR amounts, 5–15% target fraud prevalence, state/city/tier baselines, policy duration/waiting periods/co-pay, claim type, historical frequencies and provider proxies. Exact duplicates are detected and removed. The 70/15/15 stratified split is frozen. Imputation, robust scaling, one-hot encoding, smoothed target encoding and mutual-information selection are fitted on the training partition only.

Table I. Experimental protocol

Metric	Protocol
Primary selection	5-fold train CV F2
Threshold	Validation-only maximum F2
Final assessment	Untouched stratified test set
Cost proxy	FN=synthetic claim INR; FP=■3,500
Fairness	FPR/FNR/accuracy by audit groups

IV. RESULTS AND ANALYSIS

Table II. Held-out benchmark (ranked by F2)

Model	F2	Rec.	PR-AUC	AUC	ms
XGBoost	0.988	0.993	0.999	1.000	0.00
Logistic Regression (L	0.986	0.985	0.999	1.000	0.00
LightGBM	0.985	0.985	0.998	1.000	0.01
SVM (linear)	0.985	0.985	0.998	1.000	0.01
SVM (RBF)	0.984	0.996	0.997	0.999	0.05
Logistic Regression (L	0.983	0.981	0.998	1.000	0.00
Histogram Gradient Boo	0.977	0.989	0.998	0.999	0.01
Random Forest	0.976	0.978	0.997	0.999	0.02
AdaBoost	0.976	0.978	0.996	0.999	0.03
Shallow Neural Network	0.963	0.959	0.997	0.999	0.00
Decision Tree	0.946	0.963	0.937	0.980	0.00
K-Nearest Neighbors	0.928	0.929	0.947	0.970	0.05
Quadratic Discriminant	0.891	0.914	0.896	0.985	0.00
Gaussian Naive Bayes	0.862	0.900	0.876	0.979	0.00
Multinomial Naive Baye	0.809	0.888	0.821	0.951	0.00

Table III. Source adequacy decision

Source criterion	Observed status
Rows	4,500; below the ≥10,000 study criterion
Label	Binary ClaimLegitimacy field is present
Indian context	Generic locations/identifiers; not adequate
Policy/history fields	Required waiting/co-pay/history fields absent
Decision	Preserve source; use labelled synthetic fallback

Table IV. Feature families and leakage controls

Feature family	Examples / controls
Policy	sum insured, premium, co-pay, waiting period
Claim	INR amount, type, treatment, duration, procedure count
History	frequency, time since prior claim, amount baseline
Provider/context	tier, network status, distance, regional baseline
Engineered	cost deviation, utilisation, amount/day; train-only encoding

Table V. Illustrative operational cost assumptions

Error outcome	Illustrative handling
False negative	Synthetic fraudulent claim recommended approve; cost equals its INI
False positive	Legitimate claim routed to review; ■3,500 friction/review proxy

Threshold	Selected on validation maximum F2, never using test labels
Decision authority	Human reviewer verifies documents, policy and medical evidence

The F2-leading synthetic model is XGBoost. Its threshold of 0.295 was selected from validation probabilities, avoiding a default 0.50 assumption. Precision, recall, PR-AUC, MCC, calibration and INR proxy cost must be read together; a higher F2 alone does not justify automatic rejection.

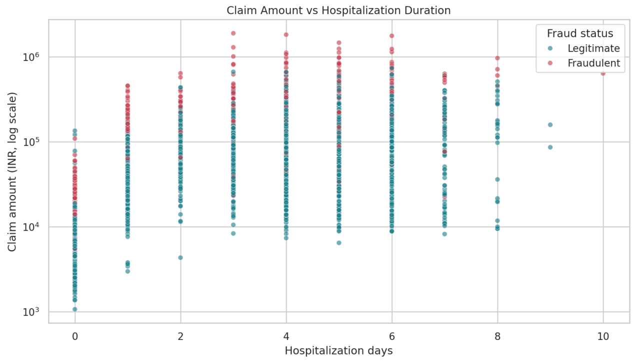

Fig. 1. Synthetic claim amount distribution by class.

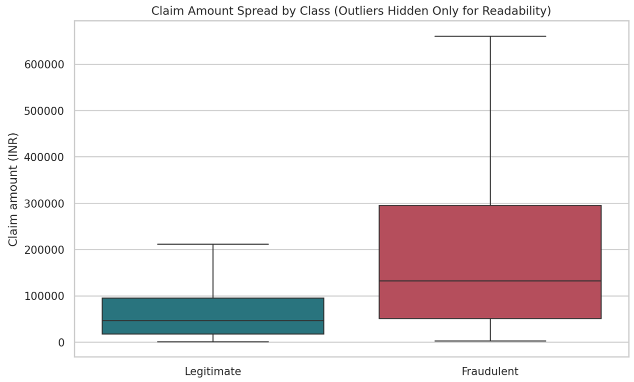

Fig. 2. Numeric correlation structure.

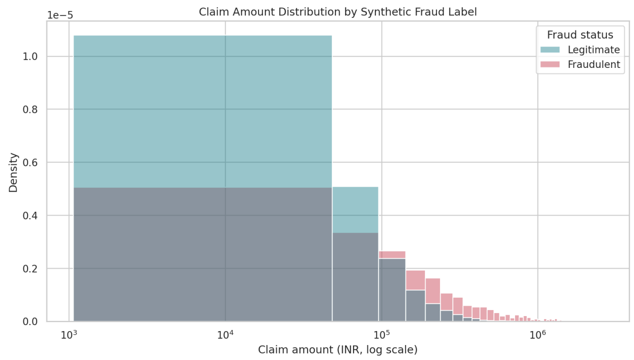

Fig. 3. Held-out all-model metric comparison.

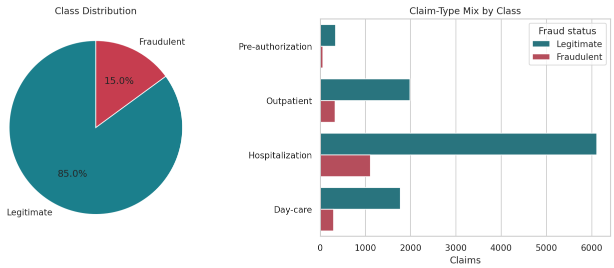

Fig. 4. Held-out ROC comparison.

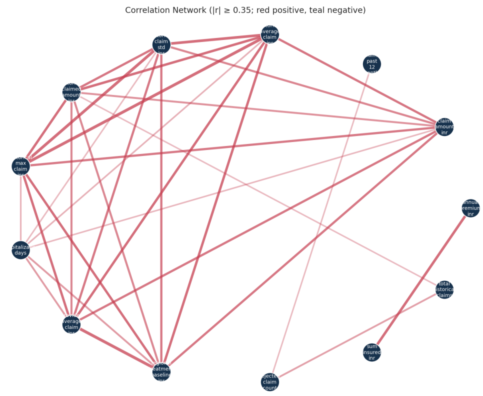

Fig. 5. Held-out precision–recall comparison.

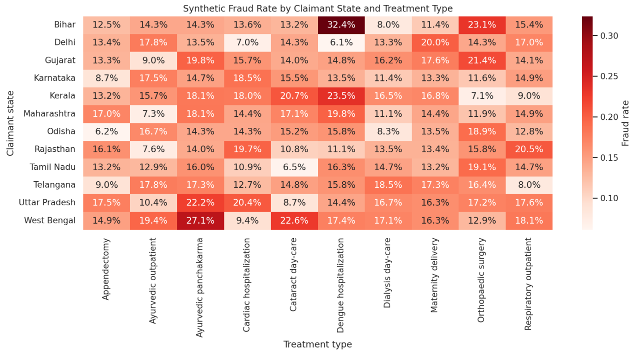

Fig. 6. Fairness/calibration audit.

V. FAIRNESS, EXPLAINABILITY AND DISCUSSION

The held-out audit reports group accuracy, FPR, FNR, precision, recall and selection rate across gender, age, geography, income and treatment. Small groups are marked rather than overinterpreted. A material disparity triggers data-quality review and a mitigation comparison; protected attributes must never act as unreviewed grounds for a denial. Native feature importances and coefficients are associations within the synthetic generator. They must be translated into reviewable evidence, such as a bill discrepancy or policy clause, before communicating with a policyholder.

VI. LIMITATIONS AND FUTURE WORK

Synthetic labels and prices prevent operational claims. Future work requires governed de-identified insurer data, temporal/entity-disjoint validation, data drift monitoring, external calibration, policy-specific cost matrices, documented appeals and human-review performance measurement. Deep-learning models should use the same frozen test protocol; the agentic workflow should add document/RAG evidence without representing LLM reasoning as factual proof.

VII. CONCLUSION

This work provides a reproducible and transparent classical baseline for educational Indian-context claim fraud screening. It makes data limitations visible, prioritizes recall-aware evidence, and packages performance with cost, interpretability and fairness audits. It is a foundation for accountable research, not an automatic insurance decision system.

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